

X-ray Multi-Classified Preprocessing, Visualization, and Model Design

Nick Calabro, Anthony Klimas, Luke MacVicar, Hilary Jaen Rodriguez

Data Overview

- Source: NIH Chest X-ray 14
- Originally 1024 x 1024 or higher
- 112,120 images
- We aim to achieve competitive accuracy (within 5%) of modern models on this problem
- Additional metadata:
 - Age
 - Gender
 - Number of followup doctor visits
 - Chest X-ray orientation
 - Original resolution



EDA Insights

- True Count Mean: 5798.286
- Standard Deviation: 5429.592
- Imbalance is clearly quite large with a range of 19,667



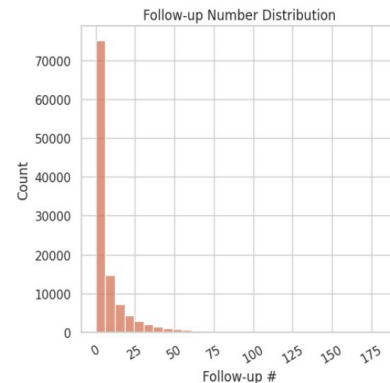
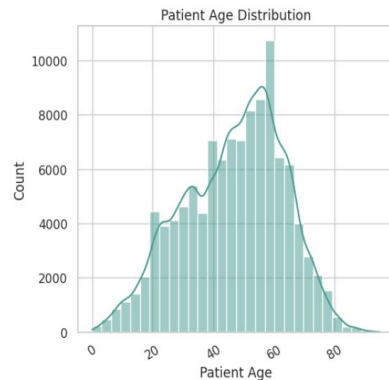
EDA Insights

- Image resolution is not standardized
 - Often too high
- Depending on training time, we'll experiment with resolution size
- Baseline will use a consistent 224 x 224 resolution
- Likely diminishing returns in accuracy as the resolution increases
- Early training cycles starting with small resolution may detect broader features
 - Increase resolution as training epochs increase



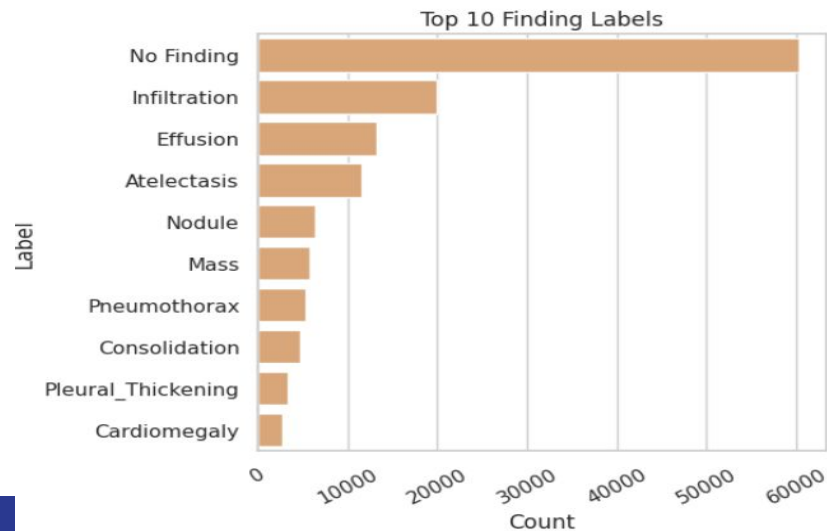
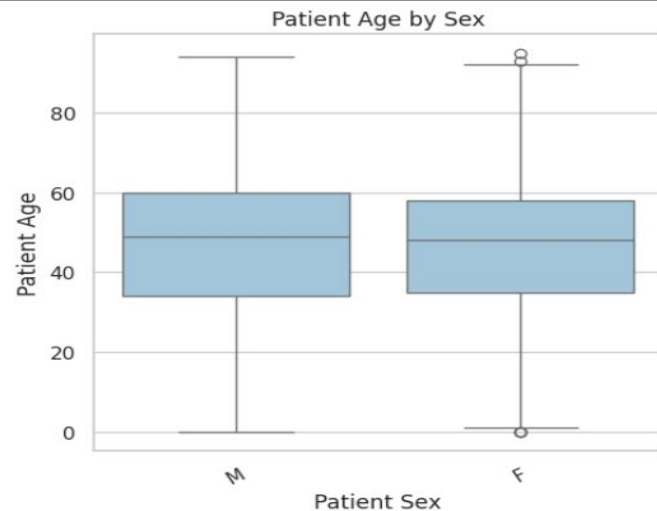
Data Visualizations

- Patient Age Distribution: This shows the ages of each patient involved in the dataset. We see that most fall between the ages of 50 and 60 years old
- Follow-up Number Distribution: Shows how many of these images are from follow ups. Which is a very small number meaning most of these are from unique patients.



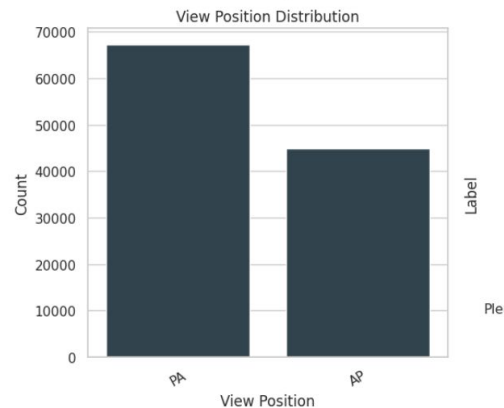
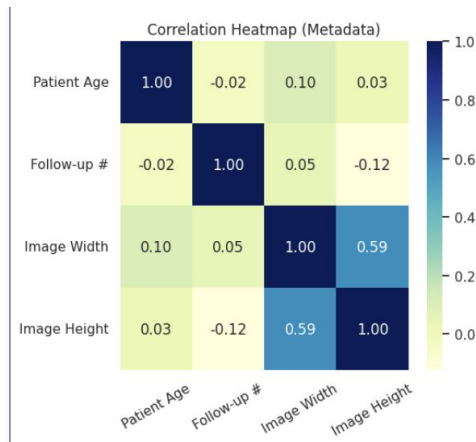
Data Visualizations

- Patient Age by sex shows the distribution of the ages for the patients distinguished by sex
- Top 10 findings shows what the distribution of findings across the data set



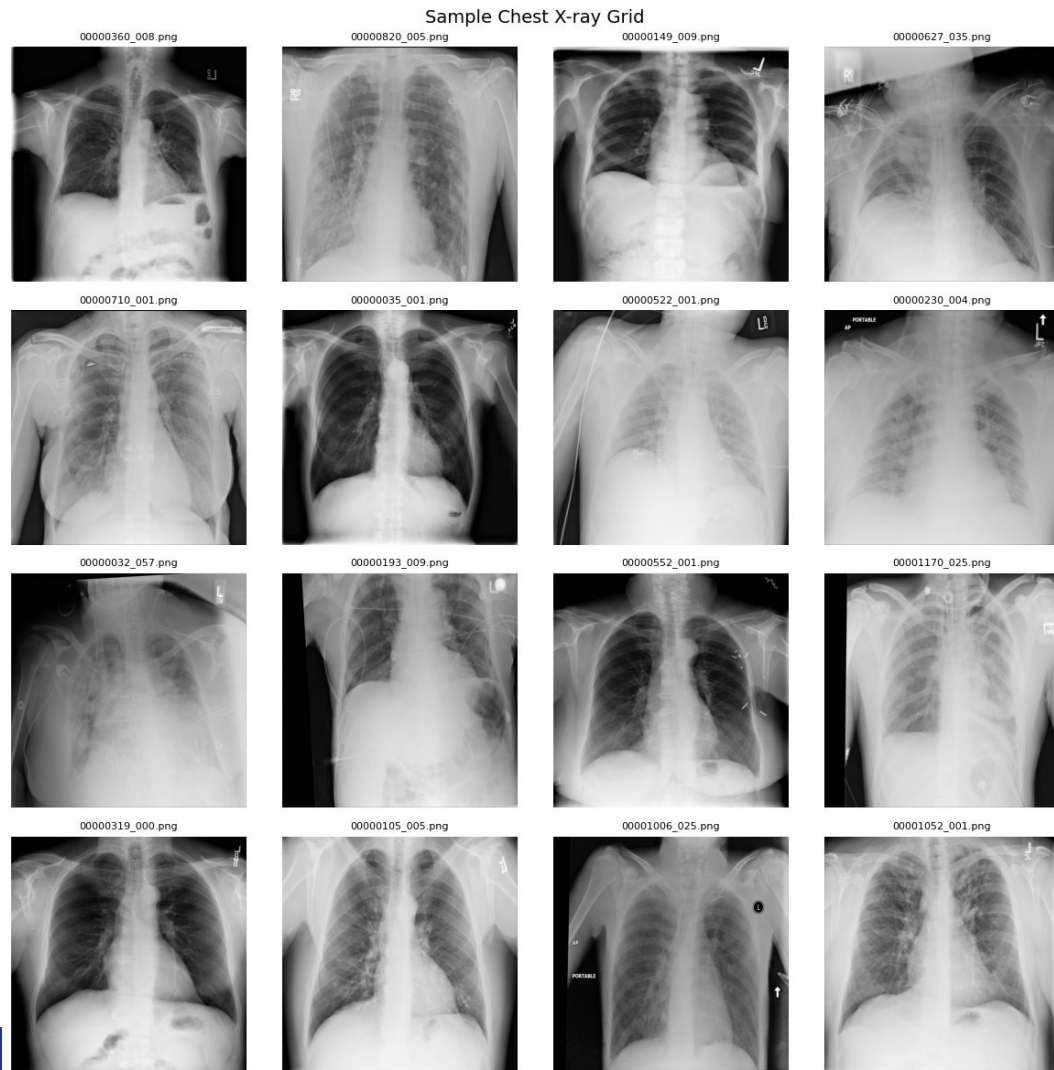
Data Visualizations

- Correlation Heatmap shows how related these aspects of the data are.
- The view position distribution is demonstrating which images are from each xray position



X-ray Grid

- This is the raw preprocessed data
- It is important to see and understand the data we are working with before putting it into the CNN



Challenges

- **Class Imbalance:**
 - Some disease appear much more frequently than others, which can cause the model to favor common classes and perform poorly on rare ones
- **Multi-Label Complexity**
 - Each X-ray image can contain multiple diseases, requiring the model to predict several labels at once instead of just one.
- **Label noise**
 - The dataset labels were generated automatically, which may introduce errors and reduce overall data quality.



Next Steps

Data preprocessing (resize, normalize):

We will preprocess the data by resizing images to a standard size and normalizing pixel values to improve model training efficiency.

Data augmentation (flip, rotate):

We will apply data augmentation techniques such as flipping and rotating images to increase data diversity and improve generalization.

Handle imbalance (class weights):

To address class imbalance, we will use techniques such as class weighting to ensure the model does not favor more frequent diseases.

Model training (Swin Transformer):

We will train a Swin Transformer model to effectively capture both local and global patterns in the images for improved classification performance.

